

Lesion Atlas

A daily skin-capture pipeline that finds acne lesions across six facial regions, holds each lesion's identity across days in a fixed anatomical coordinate frame, and turns months of that into a picture of how your skin is actually changing.

CAPTURE phone browser → local server

POSES frontal + L/R 60°

REGIONS 6

ZOOM none

LABELS 300–600 images

FALLBACK laptop webcam

THE DESIGN

Three poses at sixty degrees

Every session is three shots, no zooming: **frontal**, **left 60°**, **right 60°**, each framed to fill the frame with your face.

The angle is doing specific work. Skin at angle θ to the sensor projects at $\cos(\theta)$ of its true size, and the incidence angle for a point on your face is roughly $|\varphi - \text{yaw}|$, where φ is that point's azimuth from straight-ahead — nose at 0°, directly lateral at 90°. In a frontal shot the skin by your sideburns is nearly edge-on:

REGION	Φ	FRONTAL	45° YAW	60° YAW	90° YAW
Forehead, center	~5°	5°	40°	55°	85°
Medial cheek	~30°	30°	15°	30°	60°
Lateral cheek	~60°	60°	15°	0°	30°
Preauricular / sideburn	~80°	80°	35°	20°	10°
Incidence angle in each column. Lower is better. A 3 mm papule at the sideburn projects to about 0.5 mm frontally and 2.8 mm at 60° yaw.					

Frontally, a lesion by your sideburn occupies half a millimetre of projected width, smeared over a handful of pixels and lit at grazing incidence so it reads as either a blown specular strip or shadow. It is not recoverable — and **zooming does not fix it**, because a tighter crop shows the same useless angle, larger. Turning your head does fix it.

WHY 60° AND NOT 90°

A three-quarter angle isn't a compromise between frontal and profile — it roughly bisects the span of face you need to cover on one side, so everything from mid-cheek to sideburn lands at workable incidence at once. A full profile optimizes the sideburn alone and costs you three things:

- **Landmarks.** MediaPipe Face Mesh holds to roughly $\pm 60^\circ$ yaw. Past that, half the landmarks self-occlude and you're back to landmark-free registration — the hardest version of this problem.
- **The overlap.** At 60° , the band around $\phi \approx 30\text{--}50^\circ$ (mid-cheek) is well-seen in both the frontal and the turned pose. That shared skin is what ties the three pose coordinate frames into one head, and it gives you a registration-error estimate that keeps reporting itself every single day. Frontal and profile share almost nothing.
- **The medial cheek,** which lands at 60° incidence in a profile — worse than the frontal already gives you.

One real advantage of a profile: less depth variation than a three-quarter, so depth of field is easier. Not enough to outweigh the above.

SIX REGIONS, EACH COUNTED IN EXACTLY ONE POSE

Temple and sideburn are split out from cheek rather than folded into it — buried inside "cheek," the area you care about most can never be charted on its own. Because the poses overlap, every region needs an **owner**; a detection in a pose that doesn't own that region is kept but flagged validation-only, or you double-count everything in the overlap band.

POSE	OWNS — COUNTED HERE	ALSO VISIBLE — VALIDATION ONLY
Frontal	forehead · chin & perioral	medial cheeks, nose
Left 60°	left cheek (to the jawline) · left temple & sideburn	mid-cheek band, left forehead edge
Right 60°	right cheek (to the jawline) · right temple & sideburn	mid-cheek band, right forehead edge

SHOOTING NOTES

- **Shoot from farther on the 2× lens rather than close on the main lens.** Same framing, more depth of field, much less perspective distortion. Depth of field is a genuine constraint in a turned pose — your nose and ear sit ~10 cm apart in depth and a phone at 30 cm cannot hold both sharp. The tele sensor is often lower quality, so measure this rather than assume it.
- **Fix the yaw physically.** Floor tape for where you stand, plus a mark on the wall to each side to turn toward. "About 60°" drifting week to week reintroduces exactly the variance you're eliminating.
- **Tap to focus on the region that pose owns,** not the center of the frame.

IF 60° STILL UNDER-COVERS THE SIDEBURN

Add full 90° profiles as a fourth and fifth pose owning *only* the temple and sideburn regions — don't replace the three-quarters. Same for a chin-up shot if the underjaw turns out to matter. Add either only on evidence from real photos, never preemptively: each extra pose is daily friction, and friction is the top risk below.

Where this fails

RISK	SEV	WHY IT KILLS THE PROJECT	MITIGATION
You stop shooting	● High	Everything downstream needs 60+ consecutive days. A three-week gap is unrecoverable. This is the most likely failure mode by a wide margin.	Capture under 60 seconds, start to finish. Ghost overlay, no typing, no menus. Optimize this before anything else.
Lighting drift	● High	Invisible until month three, when your chart shows a step change on the day you replaced a lightbulb. Redness is the signal; lighting is a confounder that fully swamps it.	Printed color card in every frame, one fixed lamp, same time of day. Normalize per-frame against the card.
Registration drifts across days	● Med	Without a reliable mapping into a fixed anatomical frame there is no lesion identity, and half of what you want is gone.	Phase 0 spike before anything else. Landmarks for coarse alignment, feature matching to a per-pose atlas for the fine fit, hard rejection of bad registrations, daily cross-pose check.
Random train/test split	● Med	Consecutive days of the same face are near-duplicates. A random split puts Tuesday in train and Wednesday in test; your metrics will look excellent and mean nothing.	Split by date. Hold out contiguous blocks of days, never individual images.
Inconsistent labeling	● Med	If you can't reproduce your own labels, the model's ceiling is your noise floor — and you'll misread label noise as model error for weeks.	Write the rubric before labeling. Measure self-agreement at image 100. Under ~0.75, fix the rubric, not the model.
Scale drift	● Low	Standing 5 cm closer makes everything 15% bigger. Size thresholds in pixels become meaningless and match gates misfire.	The fiducial that fixes lighting fixes this too. Work in millimetres, never pixels.

THE BUILD • ROUGHLY 8 WEEKS AT 8–12 HRS/WEEK

Eight phases

ON SEQUENCING

The detector is in v1, but it can't be built first — it needs a few hundred labeled photos of your face that don't exist yet. So capture starts immediately (every day you delay is a day permanently missing from your longitudinal data), labeling runs alongside it, and the detector lands around week 5. Registration comes before all of it because a perfect detector with broken registration gives you daily counts and no lesion identity, which is half of what you asked for.

Spike — go / no-go

Don't build the app yet. Prove the load-bearing things work, with throwaway code and throwaway photos. Three questions.

WHAT YOU'RE SHOOTING HERE

This is calibration, not a capture session. You shoot **one pose — your left 60° — five separate times across three days**, setting up from scratch each time. One pose, five repeats. Repeatability is the whole point: one shot tells you nothing about whether the pose lands in the same place tomorrow. *The daily session, from phase 1 onward, is three shots: frontal, left 60°, right 60°.*

- **Does Face Mesh hold at 60°?** MediaPipe must fire on all five repeats with stable landmarks. If it drops out, step down to 50° — that constraint sets the angle, not aesthetics. Everything downstream assumes landmarks are available.
- **Does fine registration land?** Freeze one of the five as the **atlas** and register the other four to it. Coarse-align with landmarks, then refine by feature matching: ORB/SIFT + RANSAC → LightGlue with DISK or SuperPoint → LoFTR (detector-free, the one that works on low-texture skin). Your LOQ's GPU makes all three fast enough.
- **Which lens?** Same pose on the main lens at ~30 cm and the 2× at ~55 cm. Compare sharpness at the sideburn and at the nose *within the same frame* — depth of field is what you're testing.
- Measure honestly: mark 8–10 corresponding points by hand (moles, freckles, scars — excellent anchors and your ground truth), apply the estimated transform, report median and p95 reprojection error **in millimetres**.
- Homography assumes a plane; a turned face very much isn't one. Expect structured residuals toward the frame edges and step up to a thin-plate spline fit on the matched points.

GATE — median error under 1.5 mm, p95 under 3 mm. Worse, and you cannot distinguish adjacent lesions; reduce yaw and retry before changing anything else.

Capture rig and app

Two halves, and the physical half matters more.

THE RIG

- One diffused lamp, fixed position. Blinds shut, overheads off. Same time of day — skin erythema tracks temperature, exercise and time since showering, all of which track the clock.
- Print an ArUco marker plus a small color patch on matte paper, taped to a headband or a stand in frame. One piece of cardstock buys you **absolute mm scale, white-balance normalization, and a registration anchor**. Highest leverage per dollar in the project.
- Floor tape for your feet; wall marks left and right to turn toward.

THE APP

- FastAPI on the laptop serving a static page. Self-signed cert — `getUserMedia` needs a secure context and `localhost` doesn't help you on the phone. `mkcert`, ten minutes, once.
- Three-step wizard: frontal → left 60° → right 60°, one tap each, auto-advance, then a covariate row. No text entry.
- **Ghost overlay**. Composite the previous session's frame for that pose over the live camera at ~35% opacity, and align to it by eye — including the yaw angle, which is otherwise the hardest thing to reproduce. This is the highest-value feature in the app: it moves the registration problem to capture time, where a human solves it for free.
- Tap-to-focus defaulted to the region that pose owns.
- Request `exposureMode: 'manual'`, fixed white balance and torch via `applyConstraints`. Android Chrome honors some of this; iOS Safari essentially none. Log what was actually granted. This gap is the real argument for the eventual native app.
- Capture at full sensor resolution via `ImageCapture.takePhoto()`, not a canvas grab off the preview stream — the preview is downsampled.
- POST to the server; write `data/raw/YYYY-MM-DD/<pose>.jpg` plus a sidecar JSON with device, lens, granted constraints and timestamp.
- Webcam fallback: a `cv2.VideoCapture` script writing to the same paths, tagged with `device` so you can exclude or separately calibrate those sessions. Never silently mix them.

| GATE — a full session under 60 seconds, done seven days running.

Labeling protocol and tool

Runs in parallel with daily capture. This is the real cost of the project — budget 15–30 hours — and the most common place this kind of thing stalls.

WRITE THE RUBRIC FIRST

"What counts as a pimple" is a documentation problem, not an algorithm problem. Decide and write down, with example crops:

- Minimum diameter in mm — you have mm because of the fiducial. Something like 1.5.
- Must be elevated *or* erythematous. Flat and brown is a mark, not an active lesion.
- Explicit negatives: moles, pores, ingrown hairs, razor bumps, scars, shaving irritation. A named negative class teaches far more than an unlabeled region.

CLASSES

papule Red, raised, no visible head. The bread-and-butter case.	pustule Visible white or yellow center. Distinct enough to warrant its own class.
comedone Blackhead or closed whitehead. Small, clustered, hardest to label consistently.	mark Post-inflammatory hyperpigmentation. Not active — but tracking it is how you see healing.
other Mole, scar, ingrown hair. Hard negatives that stop the model inventing lesions.	

You can merge classes later. You cannot split them later without relabeling.

THE TOOL

- A small Flask/canvas page of your own (click to place, scroll to size the radius, number keys for class) or Label Studio with a rectangle config. Rolling your own is about a day and buys keyboard speed, which compounds over 500 images.
- Store center + radius in image pixels; convert to boxes at train time.
- **Active learning loop.** Label ~100 by hand, train a weak model, have it pre-label the next batch and only correct it. Three to five times faster per image, and it's what makes 500 tractable.

GATE — at image 100, re-label 20 images you labeled two weeks earlier without looking at the old labels. Agreement (matched within 2 mm) under ~0.75 means the

3

WEEK 3–4

Canonical space

Productionize the spike. Each *pose* gets a fixed coordinate frame in millimetres that every session of that pose maps into.

- **Three atlases, one per pose.** Freeze a high-quality session as each and never change them — changing an atlas invalidates every historical coordinate under it.
- Per image: Face Mesh landmarks for coarse alignment → detect fiducial, normalize color and scale → feature-match to the pose atlas → similarity, then homography, then TPS if residuals warrant → store the transform and its residual stats.
- **Reject, don't average.** Residual over threshold means flag the session unregistered and re-shoot. A silently bad registration corrupts lesion identity in ways you won't notice for months.
- Define the six region boundaries once in canonical coordinates, each tagged with its owning pose. Detections outside an owned region are kept and flagged validation-only so they never enter a count.
- **Daily cross-pose check.** A lesion in the mid-cheek band appears in both the frontal and the turned pose. Map both into a shared head frame and measure the disagreement. That number is your honest registration error — and unlike the phase-0 spike, it reports itself every single day forever.
- Sanity check: warp three different days into canonical space and blink between them. Moles should not move. If they do, stop and fix it.

Detector

Small-object detection on a low-data, single-subject problem.

- **Tile everything.** Lesions occupy a tiny fraction of the frame. Cut into 640 px tiles with ~20% overlap, train on tiles, run sliced (SAHI-style) inference and merge with NMS. This is the difference between a model that works and one that doesn't — and it's also why you don't need optical zoom: the detector never sees a full-face image anyway.
- Model: YOLOv8n / v11n or RT-DETR . Small model, small dataset; resist going bigger.
- Pretraining: look for **ACNE04** (Wu et al., ~1,450 images with lesion boxes). Verify it's still publicly available and check the license — I can't confirm its current status. If you get it, pretrain there and fine-tune on your own face; if not, COCO weights and heavier augmentation.
- Augmentation: aggressive color jitter and lighting simulation (the variation you'll actually face), mild rotation, flips. Avoid heavy geometric warps — they distort lesion morphology, which is your signal.
- Train one model across all poses, with pose as an input feature. Three per-pose models would each get a third of the data.
- **Split by date.** Contiguous held-out blocks of days. Say it out loud before writing the split function.

METRICS THAT MEAN SOMETHING HERE

P / R @ 2 mm

A detection matches a label if centers fall within 2 mm. IoU-based mAP is a poor fit for near-circular lesions.

Count MAE

Per region, per session. Your charts are built on counts, so measure counts directly.

Day-to-day Δ error

Error in the *change* between consecutive days. Trends matter more than levels, and nobody thinks to track this.

Tune the confidence threshold toward **precision**. A false positive creates a phantom lesion that enters tracking and pollutes your history; a false negative is a gap that tracking bridges anyway.

| GATE – precision above ~0.8 at recall above ~0.7 on held-out days.

Lesion identity across days

What makes this a tracker rather than a counter. In canonical space it's nearly easy — which is exactly why phase 0 came first.

- Project every owned detection into canonical mm coordinates.
- Match today's detections against **active lesions' last known positions** — not against yesterday's frame — with the Hungarian algorithm. Cost is euclidean distance, optionally plus an appearance term. Matching to lesion state rather than to the previous frame is what makes missed days harmless.
- Gate at ~4 mm. Pimples don't move; a match beyond that is wrong.
- Lifecycle: **born** once an unmatched detection recurs in 2 of 3 consecutive sessions (single-session detections are overwhelmingly false positives). **Resolved** after 3 consecutive sessions absent, backdated to the last sighting.
- Record class per observation, and you get lifecycle traces — papule → pustule → mark → gone — with durations. This is the genuinely interesting data, and no consumer app gives it to you.

LOG COVARIATES FROM DAY ONE

Sleep, stress, new products, diet flags, whether you worked out, whether you skipped washing. Ten seconds in the capture app. You cannot collect these retroactively, and in six months they're the difference between "here's a chart of my acne" and "here's what moves my acne." Log them long before you can analyze them.

Visualization

Four views, in descending order of how much you'll actually look at them.

- **Active count over time**, stacked by region, with a 7-day rolling mean overlaid. Daily counts are noisy and the raw line will mislead you.
- **Canonical density heatmap** — kernel density of lesion-*days* per region, so bright means "lesions recur here," not "one was here once." Render as small multiples by month and the distribution shift becomes directly visible.
- **Lesion swimlanes** — one horizontal bar per lesion from birth to resolution, colored by class over time. The most information-dense view you'll have.
- **Duration distribution** — how long lesions last. "Are they resolving faster?" is a sharper question than "are there fewer?" and it answers sooner.

Streamlit or a static Plotly page. Don't build a web app; you're the only user.

Where the ML gets interesting

Ordered by value, not by how cool they sound.

- **Self-supervised pretraining.** You'll have thousands of unlabeled skin images. Fine-tune DINOv2 on them and use the features as a detector backbone. Highest-value extension, and it uses the asset you'll have most of.
- **Resolution forecasting.** Given a lesion's first two days, predict whether it resolves within five. A clean supervised problem your own tracking data generates for free.
- **Covariate analysis.** Lagged regression of lesion births against your logged covariates. Not causal, but $n=1$ with dense daily sampling over months is a better dataset than most acne studies have.
- **Severity regression** instead of discrete classes — continuous erythema and elevation scores, more sensitive to change than counts.

ON GENERATIVE AI

It has essentially nothing to offer the core problem, and you should not plan around it. Two legitimate uses: a VLM as a *pre-labeling assistant* to speed up annotation (real, useful, worth doing in phase 2), and a natural-language query layer over your own database (fun, zero analytical value). Diffusion-based synthetic lesion generation is what people reach for — on a single-subject dataset it will almost certainly hurt, because the model learns your generator's artifacts rather than your skin. If you want impressive ML here, self-supervised pretraining is the answer.

REFERENCE

Schema and layout

SQLITE – THE THREE-TIER STRUCTURE IS THE IMPORTANT PART

```
sessions      id, captured_at, device, lamp_ok, covariates_json
images        id, session_id, pose, path, exif_json, device, lens,
              qa_score, reg_status, reg_median_err_mm, transform_json
detections    id, image_id, px_x, px_y, px_r, cls, score, model_version,
              canon_x_mm, canon_y_mm, canon_r_mm, region, owned
lesions       id, region, first_session_id, last_session_id,
              status, canon_x_mm, canon_y_mm
observations  id, lesion_id, detection_id, session_id, cls, severity
labels        id, image_id, px_x, px_y, px_r, cls, labeler, created_at
```

```
# detections are per-image and disposable – run a better model,
# regenerate them. lesions are the durable identity. observations
# is the join, and it's what every chart reads from.
#
# detections.owned = false means the lesion was visible in this pose
# but another pose owns that region. Kept for the daily cross-pose
# check, excluded from every count. This one flag is what stops you
# double-counting everything in the mid-cheek overlap.
```

REPO

```
lesion-atlas/
capture/      FastAPI server + static phone page + webcam fallback
core/
  calib.py    fiducial detection, color + scale normalization
  register.py landmark coarse align, atlas matching, QA gating
  regions.py  canonical region boundaries + pose ownership
detect/       tiling, train, sliced inference
track.py      hungarian matching, lifecycle state machine
db.py
label/        annotation UI
analysis/     plots, dashboard
data/
  raw/YYYY-MM-DD/{frontal,left_60,right_60}.jpg
  atlas/      frozen reference frame per pose – never edit
  labels/
models/
RUBRIC.md     what counts as a pimple, with example crops
```

Start here

- | | |
|----|---|
| 01 | Print the ArUco marker and color patch. Tape it to a headband. Set up the lamp, floor tape, and the two wall marks. |
| 02 | Shoot one pose — your left 60° — five separate times over three days, setting up from scratch each time, on both lenses. This is calibration, not a session; the daily session is three shots and starts in phase 1. |
| 03 | Confirm MediaPipe Face Mesh fires on all five repeats with stable landmarks. If it doesn't, step down to 50° and re-shoot. |
| 04 | Run the registration spike: SIFT → LightGlue → LoFTR, error measured in mm against hand-marked moles. Report median and p95. |
| 05 | Write <code>RUBRIC.md</code> while you wait. Cheapest hour in the project, and it gates all of phase 2. |

Estimates assume 8–12 hours a week. The labeling in phase 2 dominates the cost and is unglamorous, and it is the entire ML story — a model is only as good as the protocol behind its labels.

Nothing here is a diagnostic tool. A daily count that jumps around is measurement noise well before it is your skin changing, which is why every view in phase 6 shows a smoothed trend rather than a single day.